# Skin Disease Classification from Images

## PROJECT DESCRIPTION:

Skin diseases affect millions of people globally, and early detection plays a crucial role in effective treatment. However, accurate diagnosis often requires expert dermatologists, which can be time-consuming and expensive.

This project aims to build an intelligent and automated skin disease classification system using deep learning, particularly Convolutional Neural Networks (CNNs). The system processes dermoscopic images to classify skin conditions such as eczema, melanoma, acne, and psoriasis. It enables both medical professionals and general users to detect skin issues early by uploading an image and receiving a real-time diagnostic prediction.

By training on a large dataset of labeled dermoscopic images, the model learns patterns and features unique to each condition, improving accuracy and assisting in medical decision-making.

## FEATURES:

- **Multi-class Classification:** Capable of identifying several skin conditions with a single model.
- **Real-time Prediction:** Instant results after uploading a skin image.
- **User Interface (optional):** A simple frontend where users can upload photos for analysis.
- **Scalable Architecture:** Can be extended to include more diseases or mobile app integration.
- **Continuous Learning:** Easily updatable with new datasets to enhance performance.
- **Privacy-Focused:** Ensures secure handling of personal medical images.

**TECHNOLOGIES USED:**

- **Programming Language:** Python
- **Deep Learning Frameworks:**
  - TensorFlow – for building and training CNN models
  - Keras – for high-level neural network implementation
- **Image Processing:**
  - OpenCV – for pre-processing and enhancing image quality
- **Development Tools:**
  - Jupyter Notebook / Google Colab – for model development and experimentation
  - Matplotlib/Seaborn – for visualizing training progress and confusion matrices
- **Dataset:**

  **Kaggle DS: [https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000](https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000)**

  - **HAM10000** (Human Against Machine with 10000 training images) – a publicly available benchmark dataset for skin lesion analysis.

**OUTCOMES:**

- **Trained Deep Learning Model:** A well-performing CNN capable of classifying skin diseases with measurable accuracy and confidence scores.
- **Prototype Application:** A working tool where users can upload dermoscopic images and get disease predictions.
- **Clinical Relevance:** Demonstrates the potential of AI in healthcare diagnostics, assisting doctors in early-stage detection.
- **Research Potential:** Opens up opportunities for academic research and real-world deployment in telemedicine.

- **Performance Metrics:** Accuracy, precision, recall, and F1-score are calculated to validate model performance.